In [2]:
```
%pip install pandas numpy matplotlib seaborn
```

```
Requirement already satisfied: pandas in .\.venv\Lib\site-packages (3.1.0)
Requirement already satisfied: numpy in .\.venv\Lib\site-packages (2.5.4)
Requirement already satisfied: matplotlib in .\.venv\Lib\site-packages (3.11.2)
Requirement already satisfied: seaborn in .\.venv\Lib\site-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.9.0 in .\.venv\Lib\site-packages
(from pandas) (4.0.0)
Requirement already satisfied: tzdata in .\.venv\Lib\site-packages (from pandas) (20
26.6)
Requirement already satisfied: contourpy>=1.0.1 in .\.venv\Lib\site-packages (from m
atplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in .\.venv\Lib\site-packages (from matpl
otlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in .\.venv\Lib\site-packages (from
matplotlib) (4.68.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\.venv\Lib\site-packages (from
matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in .\.venv\Lib\site-packages (from ma
tplotlib) (26.3)
Requirement already satisfied: pillow>=9 in .\.venv\Lib\site-packages (from matplotl
ib) (13.0.0)
Requirement already satisfied: pyparsing>=3 in .\.venv\Lib\site-packages (from matpl
otlib) (3.3.3)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully")
```

```
Libraries imported successfully
```

In [9]:
```
data = {
    'Order_ID': [
        1001, 1002, 1003, 1004, 1005,
        1006, 1007, 1008, 1009, 1010,
        1011, 1012, 1013, 1014, 1015,
        1016, 1017, 1018, 1019, 1019
    ],

    'Customer_ID': [
        501, 502, 503, 504, 505,
        506, 507, 508, 509, 510,
        511, 512, 513, 514, 515,
        516, 517, 518, 519, 519
    ],

    'Customer_Name': [
        'Rahul', 'Priya', np.nan, 'Sneha', 'Kiran',
        'Divya', 'Rohit', 'Ananya', 'Vikram', 'Meena',
        'Sai', 'Pooja', 'Varun', 'Neha', 'Ravi',
        'Kavya', 'Manoj', 'Asha', 'Tarun', 'Tarun'
```

```
    ],

    'Product': [
        'Laptop', 'Mobile', 'Chair', 'Table', 'Headphones',
        'Notebook', 'Printer', 'Pen', 'Sofa', 'Monitor',
        'Desk', 'Keyboard', 'Mouse', 'Marker', 'Bookshelf',
        'Tablet', 'Smartwatch', 'Camera', 'Speaker', 'Speaker'
    ],

    'Category': [
        'Electronics', 'electronics', 'Furniture', 'Furniture',
        'Electronics', 'Stationery', 'Electronics', 'Stationery',
        'Furniture', 'Electronics', 'Furniture', 'Electronics',
        'Electronics', 'Stationery', 'Furniture', 'Electronics',
        'Electronics', 'Electronics', 'Electronics', 'Electronics'
    ],

    'Region': [
        'South', 'North', 'East', 'West', 'South',
        'south', 'East', 'West', 'North', 'North',
        'East', 'West', np.nan, 'North', 'West',
        'South', 'East', 'North', 'West', 'West'
    ],

    'City': [
        'Hyderabad', 'Delhi', 'Kolkata', 'Mumbai', 'Chennai',
        'Pune', 'Hyderabad', 'Delhi', 'Chennai', 'Mumbai',
        'Pune', 'Kolkata', 'Hyderabad', 'Delhi', 'Chennai',
        'Bangalore', 'Mumbai', 'Pune', 'Kolkata', 'Kolkata'
    ],

    'Order_Date': pd.date_range('2026-01-01', periods=20),

    'Quantity': [
        2, 3, 5, 2, 4, 10, 1, 20, 1, 2,
        3, 5, 6, 15, 2, 3, 4, 2, 5, 5
    ],

    'Unit_Price': [
        55000, 20000, 2500, 8000, 1500,
        50, 12000, np.nan, 25000, 10000,
        6000, 800, 500, 30, 7000,
        18000, 5000, 30000, 2500, 2500
    ],

    'Sales': [
        110000, 60000, 12500, 16000, 6000,
        500, 12000, 200, 25000, 20000,
        18000, 4000, 3000, 450, 14000,
        54000, 20000, 60000, 12500, 12500
    ],

    'Discount': [
        5000, 3000, 500, 1000, 200,
        0, 1000, 0, 2000, 1500,
        1000, 300, 200, 0, 700,
```

```
        2500, 1000, 3000, 500, 500
    ],

    'Profit': [
        12000, 7500, 2500, 3200, 1200,
        100, 1800, 50, 5000, 3000,
        3600, 800, 600, 90, 2800,
        6500, 4000, 8000, 2500, 2500
    ],

    'Payment_Mode': [
        'UPI', 'Credit Card', 'Cash', 'UPI', 'Debit Card',
        'Cash', 'Credit Card', 'UPI', 'Debit Card', 'UPI',
        'Cash', 'Credit Card', 'UPI', 'Cash', 'Debit Card',
        'UPI', 'Credit Card', 'Cash', 'UPI', 'UPI'
    ],

    'Shipping_Mode': [
        'Express', 'Standard', 'Standard', 'Express', 'Same Day',
        'Standard', 'Express', 'Standard', 'Same Day', 'Express',
        'Standard', 'Express', 'Standard', 'Same Day', 'Standard',
        'Express', 'Standard', 'Same Day', 'Express', 'Express'
    ],

    'Sales_Channel': [
        'Online', 'Retail', 'Online', 'Retail', 'Online',
        'Retail', 'Online', 'Retail', 'Online', 'Retail',
        'Online', 'Retail', 'Online', 'Retail', 'Online',
        'Retail', 'Online', 'Retail', 'Online', 'Online'
    ],

    'Customer_Age': [
        25, 32, 28, 35, 24, 40, 29, 22, 45, 31,
        27, 36, 26, 30, 42, 23, 34, 150, 29, 29
    ],

    'Customer_Gender': [
        'Male', 'Female', 'Male', 'Female', 'Male',
        'Female', 'Male', 'Female', 'Male', 'Female',
        'Male', 'Female', 'Male', 'Female', 'Male',
        'Female', 'Male', 'Female', 'Male', 'Male'
    ],

    'Order_Status': [
        'Delivered', 'Delivered', 'Pending', 'Delivered',
        'Shipped', 'Delivered', 'Cancelled', 'Delivered',
        'Shipped', 'Delivered', 'Pending', 'Delivered',
        'Shipped', 'Delivered', 'Pending', 'Delivered',
        'Shipped', 'Unknown', 'Cancelled', 'Cancelled'
    ],

    'Rating': [
        5, 4, 3, 5, 4, 3, 2, 5, 4, 5,
        3, 4, np.nan, 2, 4, 5, 4, 3, 2, 2
    ]
}
```

```
df = pd.DataFrame(data)

df
```

Out[9]:

	Order_ID	Customer_ID	Customer_Name	Product	Category	Region	City
0	1001	501	Rahul	Laptop	Electronics	South	Hyderabad
1	1002	502	Priya	Mobile	electronics	North	Delhi
2	1003	503	NaN	Chair	Furniture	East	Kolkata
3	1004	504	Sneha	Table	Furniture	West	Mumbai
4	1005	505	Kiran	Headphones	Electronics	South	Chennai
5	1006	506	Divya	Notebook	Stationery	south	Pune
6	1007	507	Rohit	Printer	Electronics	East	Hyderabad
7	1008	508	Ananya	Pen	Stationery	West	Delhi
8	1009	509	Vikram	Sofa	Furniture	North	Chennai
9	1010	510	Meena	Monitor	Electronics	North	Mumbai
10	1011	511	Sai	Desk	Furniture	East	Pune
11	1012	512	Pooja	Keyboard	Electronics	West	Kolkata
12	1013	513	Varun	Mouse	Electronics	NaN	Hyderabad
13	1014	514	Neha	Marker	Stationery	North	Delhi
14	1015	515	Ravi	Bookshelf	Furniture	West	Chennai
15	1016	516	Kavya	Tablet	Electronics	South	Bangalore
16	1017	517	Manoj	Smartwatch	Electronics	East	Mumbai
17	1018	518	Asha	Camera	Electronics	North	Pune
18	1019	519	Tarun	Speaker	Electronics	West	Kolkata
19	1019	519	Tarun	Speaker	Electronics	West	Kolkata

In [10]:
```
df.shape
```

Out[10]:
```
(20, 20)
```

In [11]:
```
df.isnull()
```

Out[11]:

	Order_ID	Customer_ID	Customer_Name	Product	Category	Region	City	Order_Date
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	True	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False
5	False	False	False	False	False	False	False	False
6	False	False	False	False	False	False	False	False
7	False	False	False	False	False	False	False	False
8	False	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False	False
10	False	False	False	False	False	False	False	False
11	False	False	False	False	False	False	False	False
12	False	False	False	False	False	True	False	False
13	False	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False	False
15	False	False	False	False	False	False	False	False
16	False	False	False	False	False	False	False	False
17	False	False	False	False	False	False	False	False
18	False	False	False	False	False	False	False	False
19	False	False	False	False	False	False	False	False

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
Order_ID          0
Customer_ID       0
Customer_Name     1
Product           0
Category          0
Region            1
City              0
Order_Date        0
Quantity          0
Unit_Price        1
Sales             0
Discount          0
Profit            0
Payment_Mode      0
Shipping_Mode     0
Sales_Channel     0
Customer_Age      0
Customer_Gender   0
Order_Status      0
Rating            1
dtype: int64
```

In [13]:
```
df.isnull().sum()[df.isnull().sum()>0]
```

Out[13]:
```
Customer_Name    1
Region           1
Unit_Price       1
Rating           1
dtype: int64
```

In [14]:
```
df.duplicated()
```

Out[14]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
dtype: bool
```

In [15]:
```
df.duplicated().sum()
```

Out[15]:
```
np.int64(0)
```

In [17]:
```
df=pd.concat([df,df.iloc[[0]]],ignore_index=True)
```

In [18]:
```
df.duplicated()
```

Out[18]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
20     True
dtype: bool
```

In [19]:
```
df.duplicated().sum()
```

Out[19]:
```
np.int64(1)
```

In [20]:
```
df[df.duplicated(keep=False)]
```

Out[20]:

	Order_ID	Customer_ID	Customer_Name	Product	Category	Region	City	Ord
0	1001	501	Rahul	Laptop	Electronics	South	Hyderabad	202
20	1001	501	Rahul	Laptop	Electronics	South	Hyderabad	202

In [22]:
```
df=df.drop_duplicates()
df.duplicated().sum()
```

Out[22]:
```
np.int64(0)
```

In [23]:
```
df["Sales"] = pd.to_numeric(
    df["Sales"], errors="coerce"
)

df["Sales"]
```

Out[23]:
```
0     110000
1      60000
2      12500
3      16000
4       6000
5        500
6      12000
7        200
8      25000
9      20000
10     18000
11      4000
12      3000
13       450
14     14000
15     54000
16     20000
17     60000
18     12500
19     12500
Name: Sales, dtype: int64
```

In [24]:
```
df["Sales"].max()
```

Out[24]:
```
np.int64(110000)
```

In [25]:
```
df["Sales"].min()
```

Out[25]:
```
np.int64(200)
```

In [26]:
```
df.groupby("Category")["Sales"].sum()
```

Out[26]:
```
Category
Electronics    314000
Furniture       85500
Stationery       1150
electronics     60000
Name: Sales, dtype: int64
```

In [27]:
```
df.groupby("Region")["Sales"].mean()
```

Out[27]:
```
Region
East     15625.000000
North    33090.000000
South    56666.666667
West      9866.666667
south      500.000000
Name: Sales, dtype: float64
```

In [28]:
```
df.groupby("Category")["Profit"].sum()
```

Out[28]:
```
Category
Electronics    42900
Furniture      17100
Stationery       240
electronics     7500
Name: Profit, dtype: int64
```

In [29]:
```
df.groupby("Region")["Order_ID"].count()
```

Out[29]:
```
Region
East     4
North    5
South    3
West     6
south    1
Name: Order_ID, dtype: int64
```

In [30]:
```
## Sales by Sales Channel

df.groupby("Sales_Channel")["Sales"].sum()
```

Out[30]:
```
Sales_Channel
Online    245500
Retail    215150
Name: Sales, dtype: int64
```

In [31]:
```
## Sales by Sales Channel

df.groupby("Sales_Channel")["Sales"].sum()
```

Out[31]:
```
Sales_Channel
Online    245500
Retail    215150
Name: Sales, dtype: int64
```

In [ ]: